Emad Mostafa

Data Scientist | Machine Learning Engineer
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GitHub | LinkedIn | Kaggle | Portfolio

Summary

Data Scientist with industry experience in time series, deep learning and NLP. Currently generating and validating synthetic financial data with TimeGAN and building deep learning and reinforcement learning models for trading. Part-time, I build and deploy ML features on AWS for a US pharmacy-tech company. M.Sc. student in Statistics at Cairo University.

Education

Cairo University — Cairo, Egypt
M.Sc. in Statistics (Data Science Track), Faculty of Graduate Studies for Statistical Research — *2025 – 2028 (expected)*

Helwan University — Cairo, Egypt
B.Sc. in Computer Science and Artificial Intelligence, Grade: Very Good — *2020 – 2024*

Experience

Data Scientist — Aug 2025 – Present
Limitless Labs Egypt — *Cairo, Egypt (On-site)*

- Generate synthetic financial time series with TimeGAN to augment limited market data for model training.
- Validate synthetic data with statistical tests: return distributions, autocorrelation and stylized facts.
- Build deep learning and reinforcement learning models for trading.

AI & ML Solutions Engineer (Part-Time) — Nov 2024 – Present
Banyan — *Remote, USA*

- Build and deploy AI features for a pharmacy optimization platform, from data analysis to production on AWS.
- Develop data pipelines and ML models for trend detection that support the drug recommendation engine.
- Work in a 7-person team with PostgreSQL, .NET APIs and React.

IBM Data Science Intern — Jun 2024 – Oct 2024
Digital Egypt Pioneers Initiative (DEPI) — *Hybrid*

- Built Power BI and Excel dashboards, engineered features for ML models and practiced cloud deployment.

Machine Learning Instructor — Jan 2022 – Feb 2023
IEEE FCIH Helwan University & MSP Tech Club Helwan — *Cairo, Egypt*

- Taught ML fundamentals to 100+ students at IEEE and supervised their final projects.
- Led an 8-session ML/DL workshop at MSP for 80+ attendees, with 90% positive feedback.

Technical Projects

Retail Sales Forecasting | *Prophet, XGBoost, LSTM, MLflow, Streamlit* — Code | Demo

- Compared three models with a linear baseline for daily store sales, tracked in MLflow and deployed with Streamlit.

AI Study Assistant | *Streamlit, Gemini API, OCR, Text-to-Speech* — Code

- Q&A, quizzes and flashcards from PDFs, images and YouTube videos, with Arabic and English OCR.

Invoice Automation | *Python, Selenium, Regex, Streamlit* — Code

- Automated daily e-invoice downloads from the tax portal, extracted PO numbers and emailed Excel reports.

Video Summarization | *Python, NLP, Transformers* — Code

- Summarizes long videos from their subtitles, cutting length by about 70% while keeping the key points.

Skills, Certifications & Languages

Programming: Python, SQL
ML & Deep Learning: scikit-learn, PyTorch, TensorFlow, XGBoost, GANs, LSTM, CNN, Reinforcement Learning
Time Series & Statistics: TimeGAN, Forecasting, Synthetic Data Evaluation, Hypothesis Testing
NLP & LLMs: RAG, LangChain, Transformers, NER
MLOps & Tools: AWS, Docker, MLflow, Streamlit, Git, Linux, PostgreSQL, Power BI
Certifications: ML Specialization (Stanford), TensorFlow Developer (DeepLearning.AI), IBM Data Science
Languages: Arabic (Native), English (Very Good)